Ariel Renewable Energy Limited

SOVEREIGN GREEN INFRASTRUCTURE & INDUSTRIAL DECARBONIZATION

A Member of the ArielMC Enterprise Eco-System • energy.arielmc.com

Ref: AREL-OPS-2026-MSTR-V1

Official Operational Master
Effective Date: Nov 2026

Master Operations, Sourcing & Logistics Blueprint

This master document codifies the exhaustive, engineering-grade operational architecture of **Ariel Renewable Energy Limited**. It details five core enterprise pillars: **Product & Detailed BOM**, **Logistics & SRO 159 Customs Clearance**, **Worldwide Tier-1 Sourcing**, **Technology Radar & Efficiency Optimization**, and **Turnkey Installation & Zero-Downtime RMA SOP**.

EXECUTIVE MANDATE & STATUTORY ANCHORS

Operating under the Bangladesh Power Division's Net-Metering Guidelines, SREDA accreditation, and National Board of Revenue SRO 159-Law/2026, Ariel delivers standardized C&I and residential solar infrastructure backed by 80% concessionary green refinancing (4.0%–5.5%, 10-Yr tenure with 1-Year full grace period). All engineering conforms strictly to the Bangladesh National Building Code (BNBC 2020) with basic wind resistance up to 260 km/h.

Pillar 1: Product Master & Standardized Packages

Ariel Renewable Energy deploys seven standardized, turnkey solar architectures across Industrial Factory, Commercial, and Residential sectors:

Code	Sector	Package Title	Capacity DC	Inverter AC	Roof Area	Monthly Yield	LCOE (BDT)
F1	Factory	Industrial Base-Load Offset	300 kWp	240 kW	24,000 sq ft	36,000–39,000 kWh	Tk 3.20/kWh
F2	Factory	Megawatt Flagship Net-Metering	1,000 kWp (1 MWp)	880 kW	78,000 sq ft	125,000–135,000 kWh	Tk 3.10/kWh
F3	Factory	Net-Zero Multi-Roof Heavy Industrial	2,500 kWp (2.5 MWp)	2,200 kW	200,000 sq ft	315,000–340,000 kWh	Tk 2.95/kWh
C1	Commercial	Corporate Office Tower Rooftop	75 kWp	60 kW	6,000 sq ft	9,000–10,000 kWh	Tk 3.35/kWh
C2	Commercial	Retail Mall & Hospital Net-Metering	200 kWp	160 kW	16,000 sq ft	25,000–27,000 kWh	Tk 3.20/kWh
H1	Household	Urban Villa Micro-Array (Grid-Tied)	10 kWp	10 kW	800 sq ft	1,200–1,350 kWh	Tk 3.65/kWh
H2	Household	Sovereign Villa Hybrid BESS Micro-Array	25 kWp + 30 kWh BESS	20 kW	2,200 sq ft	3,100–3,400 kWh	Tk 3.85/kWh

Itemized Bill of Materials: Package F1 (300 kWp Industrial Factory)

Item	Component	Engineering Specification	Make / Model	Qty	Warranty	Lifetime
1.01	PV Modules	620Wp N-Type TOPCon Dual-Glass Bifacial (23.0% eff, 80% bifaciality)	Jinko Tiger Neo / Longi Hi-MO 7	484 Pcs	15-Yr Product / 30-Yr Linear (87.4%)	30+ Yrs

Item	Component	Engineering Specification	Make / Model	Qty	Warranty	Lifetime
1.02	String Inverter (110kW)	110 kW AC, 98.6% eff, 10 MPPTs, Built-in AI AFCI & Type II SPD	Sungrow SG110CX-P2 / Huawei SUN2000	2 Units	5-Yr Standard + 5-Yr Extended	20 Yrs
1.03	String Inverter (40kW)	40 kW AC, 4 MPPTs, balance inverter for array symmetry	Sungrow SG40CX / Huawei SUN2000-40KTL	1 Unit	5-Yr Standard + 5-Yr Extended	20 Yrs
1.04	Mounting Structure	Anodized 6063-T6 Aluminum (15–20 µm), Non-penetrative seam clamps	Kai Aluminum / Altech (Local Extrusion)	300 kWp	15-Yr Anti-Corrosion Guarantee	30+ Yrs
1.05	Fasteners & Joinery	SUS304 Stainless Steel bolts, flange nuts, EPDM isolation pads	SUS304 / A2-70 Grade	2,800 Sets	25-Yr Anti-Rust Guarantee	30+ Yrs
1.06	Solar DC Cable	1×4mm² & 1×6mm² Tinned Copper, Cross-linked XLPO, 1500V DC	KBE Solar (Germany) / Lapp / BRB	4,200 Mtr	25-Yr Weather & UV Guarantee	30+ Yrs
1.07	AC Armored Cable	3.5 Core 185mm² / 240mm² Copper XLPE, 0.6/1kV steel tape armored	BBS Cables / BRB Cable Industries	280 Mtr	10-Yr Manufacturer Warranty	30 Yrs
1.08	DC Combiner / SPD	IP66 UV rated, Type II SPD 1500V DC, 16A DC fuses, rotary switch	Schneider Electric / Suntree / ZJBENY	6 Units	5-Yr Replacement Warranty	20 Yrs
1.09	LT Switchgear Panel	630A 4-Pole Motorized MCCB with Microprocessor Trip Unit	ABB / Schneider Electric / Siemens	1 Set	5-Yr Industrial Warranty	25 Yrs
1.10	Earthing Station	Copper-bonded electrodes (20mm×3m), 25×3mm tape, Earth R < 1Ω	Erico ERITECH / Pure Copper	1 Lot	15-Yr System Guarantee	30+ Yrs
1.11	Ariel-SaaS™ Telemetry	Class-A Pyranometer, module temp, wind sensor, 4G cloud gateway	Ariel-SaaS™ Industrial IoT Gate	1 Set	5-Yr Hardware SLA & Cloud	15 Yrs
1.12	Turnkey SREDA COD	Structural audit, 1.2m fire lanes, anti-islanding trip, 0.2S meter	Ariel Energy + SREDA Grade-A Firm	300 kWp	5-Yr Comprehensive SLA	30+ Yrs

Itemized Bill of Materials: Package F2 (1.0 MWp Megawatt Flagship)

Item	Component	Engineering Specification	Make / Model	Qty	Warranty	Lifetime
2.01	PV Modules	625Wp N-Type TOPCon Dual-Glass Bifacial (23.2% eff, 2.0+2.0mm glass)	Jinko Tiger Neo / Longi Hi-MO 7	1,600 Pcs	15-Yr Product / 30-Yr Linear (87.4%)	30+ Yrs
2.02	C&I Inverters (125kW)	125 kW AC, 12 MPPTs, Smart I-V curve diagnosis, IP66 / C5	Sungrow SG125CX-P2 / Huawei SUN2000	7 Units	5-Yr Standard + 5-Yr Extended	20 Yrs
2.03	Aluminum Structure	Anodized 6063-T6 rails (15–20 µm), BNBC 220 km/h wind certified	Kai Aluminum / Altech Extrusions	1,000 kWp	15-Yr Anti-Corrosion Guarantee	30+ Yrs
2.04	Step-Up Transformer	1,250 kVA, 0.400 / 11 kV, Dyn11, ONAN cooling, Copper Wound	Energypac Engineering / Siemens / Adex	1 Unit	5-Yr Manufacturer Warranty	30+ Yrs
2.05	11kV HT VCB Panel	630A, 25kA/3s fault rating, IDMT numerical relays, Motorized trip	Schneider Electric / ABB / Siemens	1 Panel	5-Yr Industrial Warranty	25 Yrs
2.06	Cabling Infrastructure	14,500m 1×6mm² XLPO DC + 350m 11kV 3-Core 120mm² HT Copper	KBE Solar (DC) & BRB Cables (HT)	1 Lot	25-Yr DC / 10-Yr HT Warranty	30+ Yrs
2.07	Class 0.2S Net Meter	Class 0.2S 4-quadrant bi-directional meter + dual-ratio outdoor CT/PT	Secure Meters Premier / Schneider	1 Set	5-Yr Utility Approved Warranty	20 Yrs
2.08	Enterprise SCADA	Class-A Pyranometer, optical fiber ring, drone thermography sync	Ariel-SaaS™ Enterprise Telemetry	1 System	5-Yr Telemetry SLA & Support	20 Yrs
2.09	Turnkey COD Protocol	IEB certification, RSC compliance, Anti-Islanding Trip Test	Ariel Energy + SREDA Grade-A Partner	1,000 kWp	5-Yr Workmanship & Performance	30+ Yrs

Itemized Bill of Materials: Package H2 (25 kWp + 30 kWh BESS Sovereign Villa)

Item	Component	Engineering Specification	Make / Model	Qty	Warranty	Lifetime
7.01	PV Modules	580Wp N-Type TOPCon All-Black Aesthetic Panels	Jinko Tiger Neo All-Black / Canadian Solar	44 Pcs	15-Yr Product / 30-Yr Linear	30 Yrs
7.02	Hybrid Inverter (20kW)	Three-Phase Hybrid Inverter, 10ms seamless UPS transfer switch	Deye SUN-20K-SG01HP3 / Huawei LUNA	1 Unit	5-Yr Standard + 5-Yr Extended	15–20 Yrs
7.03	LiFePO4 BESS Storage	30 kWh High-Voltage LFP Rack, 1C discharge, thermal fire suppression	CATL / BYD Battery-Box / Deye LFP	1 Rack	10-Yr / 6,000 Cycles @ 80% DoD	15 Yrs
7.04	Zero-Export Meter	Smart energy meter with CTs & dynamic peak-shaving dispatch	Eastron / Deye Smart Meter	1 Unit	5-Yr Warranty	15 Yrs
7.05	Pergola Structure	Architectural powder-coated aluminum waterproof rooftop canopy	Kai 6063-T6 Powder-Coated Truss	25 kWp	15-Yr Structural Warranty	30 Yrs

Pillar 2: Logistics & Customs Clearance Blueprint

Under National Board of Revenue (NBR) **SRO No. 159-Law/2026/14/Customs**, customs duties on primary solar hardware are reduced to 0%–1%, saving ~16% on total project CapEx:

Hardware Description	H.S. Code	Total Duty Incidence	Mandatory Compliance Rule for Customs Clearance
Assembled Solar PV Modules	**8541.43.00**	**0% – 1.0%**	Attach SREDA Online Certification & OEM IEC 61215/61730 test reports.
Solar Inverters (String/Central)	**8504.40.90**	**1.0% (AIT Only)**	LC and Commercial Invoice must state: *"Solar Photovoltaic Inverter for Renewable Power"*.
LiFePO4 Lithium BESS Storage	**8507.60.00**	**1.0%**	Must state *"Lithium-Ion Battery Storage for Solar BESS"* to prevent 37% automotive/UPS duty.
Aluminum Mounting Profiles	**7610.90.00**	**58.6% (Tax Trap)**	**DO NOT IMPORT.** Source 100% locally from Kai Aluminum / Altech in BDT to eliminate this duty.

Client ERQ USD Import Account Mechanism ($0 LC Exposure for Ariel)

For export-oriented factory clients (RMG, textiles, leather), the foreign import Letter of Credit (LC) is opened directly through the client's **Export Retention Quota (ERQ) USD account**. This achieves three vital objectives:

- **Zero Working Capital Strain on Ariel**: Eliminates 100% of Ariel's cash and bank LC limits for hardware imports.
- **Zero Forex Risk**: Transaction settles directly in USD from export proceeds, eliminating BDT/USD currency volatility.
- **Direct Client Savings**: The factory saves 2%–3% in commercial bank currency conversion fees.

Container Loading Density & CBM Math (40ft High Cube Container)

- **620W N-Type TOPCon Panels**: 36 panels/pallet × 26 vertical pallets = **936 panels per 40ft HQ = 580.32 kWp**. Gross weight: 24,500 kg (strictly within Bangladesh road weighbridge limits).
- **110 kW String Inverters**: 2 units/wooden crate × 32 crates = **64 inverters = 7.04 MW AC capacity** per container.
- **Gazipur Central Staging Warehouse**: 12,000 sq ft covered staging floor in the Konabari/Joydebpur industrial corridor. Hardware is received, barcode-scanned, and inspected via electroluminescence (EL) cameras, then dispatched just-in-time in 100 kWp batches as roof rails are mounted.

Pillar 3: Worldwide Sourcing Matrix

Ariel maintains direct relationships with BloombergNEF (BNEF) Tier-1 manufacturers that have certified local representation and fast RMA turnaround in Bangladesh:

Component	Manufacturer	Flagship Series	Tech / Grade	Efficiency	Dhaka Office Presence	Benchmark FOB Price	Standard Warranty
PV Module	**Jinko Solar** (Preferred)	Tiger Neo 72HL4-BDV	16BB N-Type TOPCon	23.23%	Liaison Office: Gulshan-1, Dhaka	$0.092 – $0.102 / Wp	15-Yr Product / 30-Yr Linear
PV Module	**Longi Solar**	Hi-MO 7 (600–625W)	HPDC Dual-Glass N-Type	23.10%	Country Manager & Tier-1 Distributors	$0.093 – $0.104 / Wp	15-Yr Product / 30-Yr Linear
Inverter	**Huawei** (Preferred)	SUN2000-100KTL-M2	10 MPPTs, Built-in AI AFCI	98.80%	Huawei Bangladesh, Gulshan-2, Dhaka	$0.038 – $0.045 / W	5-Yr Standard + 5-Yr Extended
Inverter	**Sungrow** (Preferred)	SG110CX-P2 / SG125CX	12 MPPTs, IP66 / C5 Anti-Rust	98.60%	Dedicated Service Hub: Uttara, Dhaka	$0.035 – $0.042 / W	5-Yr Standard + 5-Yr Extended
BESS Storage	**CATL**	EnerOne Liquid-Cooled LFP	LiFePO4 with Liquid Thermal	94.50%	Regional South Asia Desk	$125 – $145 / kWh	10-Yr / 8,000 Cycles @ 80%
Mounting Rails	**Kai Aluminum**	6063-T6 Solar Rail Suite	15–20 µm Anodized, Tensile ≥205MPa	N/A	Factory: Tongi Industrial Area, Gazipur	Tk 380 – Tk 420 / kg (BDT)	15-Yr Structural Anti-Corrosion
AC Cables	**BRB Cables**	11kV HT & 0.6/1kV LT Copper	99.99% Annealed Electrolytic Copper	N/A	HQ: Panthapath, Dhaka; Plant: Kushtia	Factory List minus 28%–32%	10-Yr Manufacturer Warranty

Pillar 4: Technology Benchmarking & Radar Pipeline

Why N-Type TOPCon is Superior for Bangladesh Industrial Roofs

- **Superior Temperature Coefficient (-0.29%/°C vs -0.35%/°C for PERC)**: Metal factory roofs in Gazipur and Savar reach surface temperatures of 65°C to 72°C in May. N-Type TOPCon experiences 3.5% to 5.0% lower thermal degradation, generating substantially more kilowatt-hours during extreme heat.
- **Higher Bifaciality (80%–85% vs 70% PERC)**: Generates power from reflected albedo light off white PEB roof surfaces or concrete roofs, delivering an extra 6% to 12% energy yield.
- **Zero Light-Induced Degradation (LID)**: Eliminates boron-oxygen defects, keeping Year-1 degradation strictly below 1.0%.

Continuous Technology Intelligence Radar

Ariel tracks global photovoltaic advancements through four automated channels:

- **Quarterly BNEF Tier-1 Audit**: Re-verifies tiering of module makers to maintain 100% bankability with IDCOL and scheduled banks.

- **Weekly Spot Price Tracker**: Monitors wafer, cell, and polysilicon spot pricing on PV Tech and InfoLink to time import LC openings at market lows.
- **Ariel-SaaS™ Telemetry AI**: Connects cloud pyranometers and string monitors every 60 seconds; executes automated 15-minute I-V curve scans across 1.0 MWp arrays to isolate micro-cracks and shading.

Pillar 5: Operations, Installation, O&M SLA & Warranty RMA SOP

Turnkey 5-Phase Installation SOP

1. **Phase 1: Structural Audit & Wind Uplift Testing**: IEB structural engineer certifies PEB rafters for 180 km/h (Dhaka/Gazipur) to 260 km/h (Chattogram) gale-force winds under BNBC 2020.
2. **Phase 2: Non-Penetrative Mounting**: Standing-seam aluminum clamps torqued to 14 N·m with EPDM seals (zero roof holes). Aerodynamic ballasted concrete mounts on commercial RCC slabs.
3. **Phase 3: DC Cabling & Fire Pathways**: Mandatory 1.2-meter continuous fire lanes per FSCD and RSC rules. UV-resistant HDG perforated cable trays. MC4 connectors torqued to 2.8 N·m.
4. **Phase 4: Cold Electrical Commissioning**: Megger insulation resistance test at 1,000V DC (≥50 MΩ). Station earth resistance certified to < 1.0 Ω.
5. **Phase 5: Hot Commissioning & Anti-Islanding Trip Test**: Joint inspection with DESCO/DPDC/BREB utility committee. Grid breaker pulled; inverters decouple in < 0.5 seconds. Class 0.2S smart meter installed and COD signed.

After-Sales Service Level Agreement (O&M SLA)

- **Preventative Cleaning**: Bi-weekly / monthly cleaning with demineralized water (<50 ppm TDS) before 8:00 AM.
- **Thermographic Drone Scans**: Semi-annual RTK thermal infrared imaging to detect cell micro-cracks and hotspots.
- **Guaranteed Availability**: **99.0% daytime plant availability**.
- **Mean Time to Respond (MTTR)**: Under 4 hours on-site in Dhaka/Gazipur/Savar; under 8 hours in Chattogram.

Zero-Downtime Warranty Return SOP (RMA)

1. **Fault Detection**: Ariel-SaaS™ alerts if string yield deviates by >10% from adjacent strings. Ticket logged within 60 minutes.
2. **On-Site Diagnostic**: Field engineer verifies fault with IV curve tracer and FLIR thermal camera within 24 hours.
3. **48-Hour Buffer Replacement**: Ariel maintains a **2% Spare Module Buffer and 1 Spare Inverter** at the Gazipur Staging Depot. The defective component is replaced within 48 hours from buffer stock, restoring 100% generation immediately with zero revenue loss to the client.
4. **OEM Settlement**: Ariel files the warranty claim with the OEM's local Dhaka office (Huawei in Gulshan, Sungrow in Uttara, Jinko in Gulshan), securing a new unit to restock the Gazipur buffer.

Issued & Approved by:

Syed Akramul Hassan
Managing Director & CEO
Ariel Renewable Energy Limited
A Member of the ArielMC Enterprise Eco-System

Corporate Command Hub:

House 30, Road 07, Block C, Niketon, Gulshan-1, Dhaka-1212
Primary Domain: **energy.arielmc.com**
Command Hub: **energy.arielmc.com/crm**
Official Incorporation: Reg. C-194820